BINOMIAL THEOREM

The Binomial Theorem, applied

The statement, then the steps on $(a+b)^2$.

THEOREM

For any non-negative integer n and any values a and b: $(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k$.

1. **COUNT THE TERMS**
   $n = 2$, so the expansion has $n + 1 = 3$ terms, indexed by $k = 0, 1, 2$.
2. **COEFFICIENTS**
   Row 2 of Pascal's triangle: $\binom{2}{0}, \binom{2}{1}, \binom{2}{2} = 1, 2, 1$.
3. **TERMS**
   $(a+b)^2 = a^2 + 2ab + b^2$.
4. **CHECK**
   With $a = 2$ and $b = 3$: $4 + 12 + 9 = 25 = (2+3)^2$.

DROPPING THE MIDDLE TERMS

Writing $(a+b)^2$ as $a^2 + b^2$ keeps only the two end terms. With $a = 2$ and $b = 3$ that gives $4 + 9 = 13$, but $(2+3)^2 = 25$; the missing middle term is $2ab = 12$.